Box–Cox vs Variance-Stabilizing Transformations

Motivation

Linear regression assumes:

- Linearity in the mean
- Constant variance
- Approximately normal errors

Transformations are commonly used when these assumptions fail.

Two broad approaches:

- Variance-stabilizing transformations (VSTs)
- Box–Cox transformations

Variance-Stabilizing Transformations (VSTs)

A variance-stabilizing transformation is derived from an assumed mean–variance relationship:

$$\mathrm{Var}(Y \mid \mu) = g(\mu)$$

Choose a transformation $h(Y)$ such that:

$$\mathrm{Var}(h(Y)) \approx \text{constant}$$

Key idea: - Grounded in probability theory - Tied to the data-generating distribution

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Common Variance-Stabilizing Transformations

Distribution	Typical transformation
Poisson (counts)	$\sqrt{Y}$
Binomial (proportions)	$\arcsin(\sqrt{Y})$
Gamma / multiplicative errors	$\log(Y)$
Normal	identity

When to Use a VST

Use a variance-stabilizing transformation when:

- The data-generating distribution is known or well motivated
- Variance clearly depends on the mean
- Interpretation is linked to the scientific process

Examples: - Event counts - Rates and intensities - Proportions

In many cases, a GLM is preferable to transforming the response.

Box–Cox Transformation

The Box–Cox family is:

$$Y^{(\lambda)} = \begin{cases} \dfrac{Y^{\lambda} - 1}{\lambda}, & \lambda \neq 0 \\ \log(Y), & \lambda = 0 \end{cases}$$

Properties:

- Requires strictly positive responses
- λ is estimated from the data
- Chosen by maximizing the Gaussian likelihood

What Box–Cox Tries to Fix

Box–Cox simultaneously aims to improve:

- Residual normality
- Constant variance
- Linearity of the mean function

The transformation is selected to optimize model fit, not theory.

Interpretation of λ

Common values:

- $\lambda \approx 1$: no transformation
- $\lambda \approx 0$: log transformation
- $\lambda \approx 0.5$: square-root–type transformation

Box–Cox often recovers familiar transforms in a data-driven way.

When to Use Box–Cox

Use Box–Cox when:

- The response is continuous and positive
- There is no strong prior on the distribution
- Diagnostics are broadly poor:
  - Skewed residuals
  - Mild to moderate heteroskedasticity
  - Curvature in residual plots

Common in exploratory or applied regression settings.

What Box–Cox Is Not Good For

Avoid Box–Cox when:

- The response includes zeros or negatives
- Data are discrete counts or proportions
- A GLM is clearly appropriate
- Interpretation on the original scale is critical

Relationship Between VSTs and Box–Cox

Many classical VSTs are special cases of Box–Cox:

- $\log(Y)$ corresponds to $\lambda = 0$
- $\sqrt{Y}$ corresponds to $\lambda = 0.5$

Difference in philosophy:

- VSTs: theory-driven (variance structure)
- Box–Cox: likelihood-driven (model fit)

Practical Decision Rule

1. Counts, proportions, rates:
   - Use a GLM
   - VSTs mainly for visualization
2. Continuous, positive response with unclear distribution:
   - Try Box–Cox
3. If Box–Cox suggests a familiar transform:
   - Prefer the simpler, interpretable option

Key Takeaway

Variance-stabilizing transformations come from probability theory.

Box–Cox transformations come from likelihood optimization.

Use: - VSTs when you trust the distribution - Box–Cox when diagnostics matter more than assumptions